Linux kernel sources

Location of official kernel sources

- The mainline versions of the Linux kernel, as released by Torvalds
  - These versions follow the development model of the kernel (`master` branch)
  - They may not contain the latest developments from a specific area yet
  - A good pick for products development phase
  - https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git
- The stable versions of the Linux kernel, as maintained by a maintainers group
  - These versions do not bring new features compared to Linus `tree`

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- Only bug fixes and security fixes are pulled there

- Each version is stabilized during the development period of the next
  mainline kernel

- Certain versions can be maintained for much longer, 2\$plus\$\_years

- A good pick for products commercialization phase
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Location of official kernel sources

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  - #link("https://git.kernel.org/pub/scm/linux/kernel/git/stable/linux.git")

=== Location of non-official kernel sources

- Many chip vendors supply their own kernel sources

  - Focusing on hardware support first

  - Can have a very important delta with mainline Linux

  - Sometimes they break support for other platforms/devices without
    caring

  - Useful in early phases only when mainline hasn
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t caught up yet (many vendors invest in the mainline kernel at the same time)

- Suitable for PoC, not suitable for products on the long term as usually no updates are provided to these kernels

Location of official kernel sources

- Getting stuck with a deprecated system with broken software that cannot be updated has a real cost in the end

- Many kernel sub-communities maintain their own kernel, with usually newer but fewer stable features, only for cutting-edge development
  - Architecture communities (ARM, MIPS, PowerPC, etc)
  - Device drivers communities (I2C, SPI, USB, PCI, network, etc)
  - Other communities (real-time, etc)
  - Not suitable to be used in products

Getting Linux sources

- The kernel sources are available from `https://kernel.org/pub/linux/kernel` as **full tarballs** (complete kernel sources) and **patches** (differences between two kernel versions).
- But today the entire open source community has settled in favor of Git
  - Fast, efficient with huge code bases, reliable, open source
  - Incidentally written by Torvalds

Going through Linux sources